

1. Administrative & Study Identification

Full Study Title: Study of MK-4464 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced/Metastatic Solid Tumors (MK-4464-001) **Short Title/Acronym:** MK-4464-001 **Protocol Number:** MK-4464-001 / EU CT: 2023-504855-28-00 **NCT Number:** NCT05514444 **Posting ID:** 611386539871345332 **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** MK-4464 (NK cell engager targeting CEACAM5) and pembrolizumab (Trade Name: Keytruda, ATC Code: L01FF02) **Phase of Development:** Phase 1 **Trial Status:** Completed **Medical Monitor:** Merck Clinical Trials Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, pharmacokinetics, and preliminary efficacy of MK-4464 as monotherapy and in combination with pembrolizumab in participants with advanced/metastatic solid tumors. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Progression-Free Survival (PFS), minimum and maximum plasma concentrations, and area under the plasma concentration-time curve for MK-4464, as well as to track biodistribution and tumor uptake using Zirconium-89 (⁸⁹Zr) labeled MK-4464 via PET imaging.

2.2 Background & Rationale To address the critical unmet medical need in patients with advanced/metastatic solid tumors that have failed standard therapies. MK-4464 is a novel Natural Killer (NK) cell engager that targets CEACAM5-expressing tumor cells and binds to NKG2D and CD16. Preclinical studies show it potently directs NK cells to kill cancer cells. Combining MK-4464 with the PD-1 inhibitor pembrolizumab may further enhance the immune system's cancer-fighting ability and provide a synergistic anti-tumor immune response.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm, dose-escalation (Monotherapy and Combination arms) **Blinding:** Open-label **Randomization:** N/A (Assignment based on dose-escalation cohort allocation)

3.2 Planned Enrollment & Duration **Targeted Enrollment:** 64 participants **Number of Sites:** Multicenter **Trial Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Have a histologically or cytologically confirmed advanced/metastatic solid tumor by pathology report and have received, been intolerant to, been ineligible for, or refused all treatment known to confer clinical benefit. 2. Must submit a baseline tumor sample for analysis. 3. Have a performance status of 0 or 1 on the Eastern Cooperative Oncology Group (ECOG) Performance Scale. 4. Human immunodeficiency virus (HIV) infected participants must have well controlled HIV on antiretroviral therapy (ART). 5. Participants who are HBsAg positive are eligible if they have received HBV antiviral therapy for at least 4 weeks, and have undetectable HBV viral load before randomization.

4.2 Exclusion Criteria 1. Has had chemotherapy, definitive radiation, or biological cancer therapy within 4 weeks (2 weeks for palliative radiation) before the first dose of study intervention or has not recovered to CTCAE Grade 1 or better from any AEs that were due to cancer therapeutics administered more than 4 weeks earlier. 2. Has a history of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. 3. Has clinically active central nervous system (CNS) metastases and/or carcinomatous meningitis. 4. Has an active infection requiring therapy. 5. History of an allogeneic stem cell transplant or a solid organ transplant. 6. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 7. Has an active autoimmune disease that has required systemic treatment in the past 2 years. 8. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 9. Has known psychiatric or substance abuse disorders that would interfere with the participant's ability to cooperate with the requirements of the study. 10. Has not fully recovered from any effects of major surgery without significant detectable infection. 11. Has received radiation therapy to the lung that is >30 gray (Gy) within 6 months of the first dose of study treatment. 12. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 13. Is currently participating and receiving study intervention in a study of an investigational agent or has participated and received study intervention in a study of an investigational agent or has used an investigational device within 28 days.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-4464 administered in escalating doses every 3 weeks. In combination arms, it is given with a 200 mg dose of pembrolizumab (Keytruda; ATC L01FF02) every 3 weeks. A subset of participants received ⁸⁹Zr-MK-4464 + MK-4464 on Cycle 1 Day 1 for PET imaging. **Route**

of Administration: Intravenous (IV) Infusion. **Duration of Treatment:** Each cycle is 3 weeks. Treatment continued for up to 35 cycles (approximately 2 years) or until disease progression/unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptomatic management as per institutional guidelines. **Prohibited:** Live or live-attenuated vaccines within 30 days prior to the first dose; concurrent investigational agents or devices within 28 days.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy Collection
Baseline	Cycle 1, Day 1	First IV Dose (MK-4464 monotherapy or ⁸⁹ Zr-MK-4464 for imaging arm), PET Imaging
Interim Visits	Every 3 Weeks	Cycles 2-35: MK-4464 IV $\pm$ pembrolizumab 200 mg IV, PK/PD Sampling, Safety Assessments
End of Study	Week ~105 (Cycle 35)	Final Safety Follow-up, Efficacy Evaluation, Survival Status tracking

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Number of Participants Experiencing Dose-Limiting Toxicities (DLTs):** A DLT is any toxicity assessed by the investigator to be possibly, probably, or definitely related to study intervention administration that results in a change to a given dose or a delay in initiating the next cycle. All toxicities graded using NCI CTCAE.
- 2. Number of Participants Who Experience At Least One Adverse Event (AE):** An AE is defined as any untoward medical occurrence in a participant, temporally associated with the use of study treatment, whether or not considered related to the study treatment.
- 3. Number of Participants Who Discontinue Study Treatment Due to an AE:** Discontinuations due to temporally associated untoward medical occurrences, whether or not considered related to the study treatment.

7.2 Secondary Outcomes

1. Minimum Plasma Concentration (C_{min}) of MK-4464
2. Maximum Plasma Concentration (C_{max}) of MK-4464
3. Area Under the Plasma Concentration-Time Curve (AUC) of MK-4464

(Exploratory evaluations include Objective Response Rate [ORR], Progression-Free Survival [PFS], and MK-4464 tumor uptake and biodistribution via ⁸⁹Zr-MK-4464 PET imaging).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for DLTs to guide the monotherapy dose-escalation and subsequent combination therapy escalation. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for Phase 1 oncology trials. **Data Monitoring Committee (DMC):** Internal Safety Review Committee (SRC) evaluated cohort safety data before approving dose escalations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of study medication); Efficacy Evaluable Set. **Sample Size Justification:** Standard Phase 1 dose-escalation constraints. The starting dose was determined using a Minimum Anticipated Biological Effect Level (MABEL) approach derived from preclinical cytolytic activity data. **Handling of Missing Data:** Missing pharmacokinetic and safety data were handled per standard Phase 1 statistical principles (typically no imputation for safety).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring consistent with early-phase FIH (First-in-Human) biologic trials. **Audit Trail:** Validated eCRF systems with complete audit trails, query management, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-4464-001 **Registry Number:** NCT05514444 **Posting ID:** 611386539871345332 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** Study of MK-4464 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced/Metastatic Solid Tumors **Official Regulatory Title:** A Phase 1, Open-label, Multicenter Study of the Safety, Pharmacokinetics, and Pharmacodynamics of MK-4464 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced/Metastatic Solid Tumors

12. Clinical Framework (Oncology Specific)

Disease Areas: Advanced/Metastatic Solid Tumors, Neoplasms **Semantic Type:** Neoplastic Process **Preferred UMLS Name:** Malignant Neoplasms, Metastatic **Diagnosis Threshold:** Relapsed or refractory

disease where standard life-prolonging measures are exhausted.
Medical Methodology: NK Cell Engager (Targeting CEACAM5/NKG2D/CD16) + Immune Checkpoint Inhibitor (anti-PD-1) **Optimization Target:** Evaluation of MTD, safety profile, and immune-mediated tumor clearance.

13. Study Procedures & Methodology

Management Pathway: Stepwise dose-escalation (Monotherapy evaluation preceded Combination evaluation). **Education Protocol (HCP):** Site initiation and training focusing on dose-limiting toxicities and infusion reaction management. **Education Protocol (Patient):** Comprehensive informed consent regarding the investigational nature of MK-4464 and the use of radioactive isotopes (^89Zr) for imaging. **Quality Audit Mechanism:** Centralized SRC review of clinical safety parameters and PK data before opening new dose cohorts.

14. Observation & Follow-Up Schedule

Total Duration: Up to 35 treatment cycles (approx. 2 years) plus safety/survival follow-up. **Onsite Visit Cadence:** Day 1 of every 3-week cycle. **Remote Monitoring Frequency:** Inter-cycle continuous AE reporting. **Checklist Review Interval:** End of each 21-day cycle (DLT observation period).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				